

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

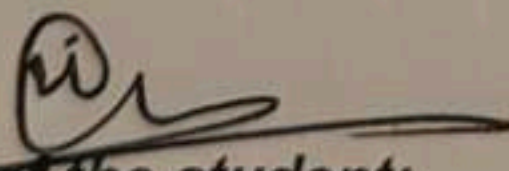
CONCEPT MAPPING

Code of Conduct:

(192324297)

I J. Vishnu Vardhan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

J. 

Signature of the student:

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

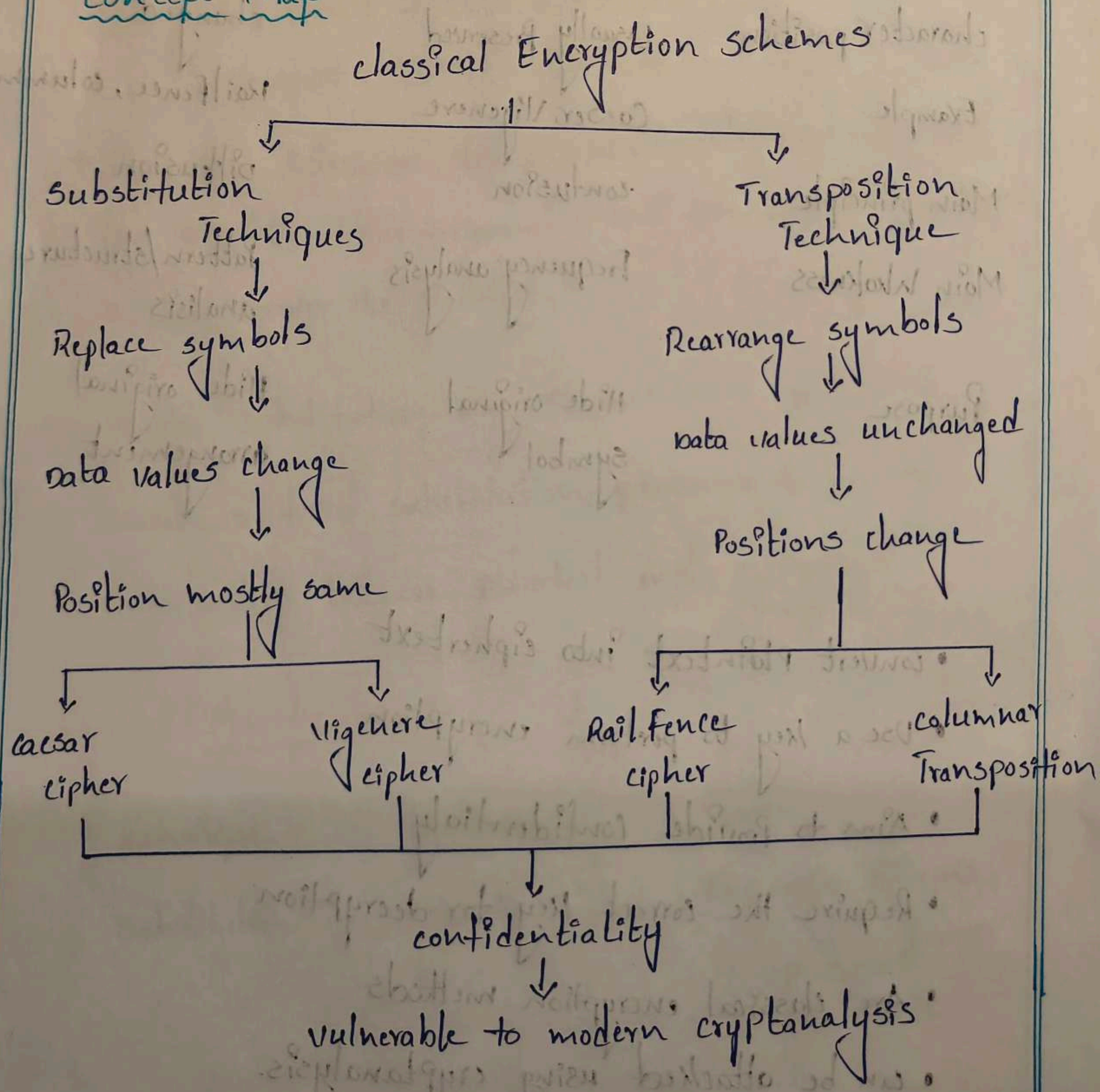
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1) Classical Encryption Schemes

To construct a concept map comparing Substitution Technique and Transposition Techniques, showing their similarities difference, effects on data structure and security level, and vulnerability to attacks.

Concept Map



Differences and similarities

Feature	Substitution	Transposition
Basic operation	Replace characters with other characters	Rearrange positions of characters
Data value	changed	unchanged
character position	Usually Preserved	changed
Example	Caesar, Vigenere	Rail Fence, columnar
Main principle	confusion	diffusion
Main weakness	Frequency analysis	Pattern/structure analysis
Purpose	Hide original symbol	Hide original arrangement

Similarities :-

- Convert Plaintext into ciphertext
- Use a key to perform encryption
- Aim to provide confidentiality
- Require the correct key for decryption
- Are classical encryption methods
- Can be attacked using cryptanalysis.

Effect on Data structure and Security level

substitution

plaintext : HELLO

ciphertext : KHOOR

The letters are replaced

data structure:

characters $\rightarrow$ changed

positions $\rightarrow$ Mostly preserved

Transposition

plaintext : HELLO

ciphertext : HLOEL

Security comparison

Substitution

changes character identity

can hide direct letter pattern

simple versions vulnerable to frequency analysis

Transposition

keep original characters

only changes their positions

letter frequencies remain unchanged

Q. Which is More Vulnerable and Why?

Transposition is generally more vulnerable to frequency based analysis when used alone.

Reason:

In transposition, the actual characters are not changed.

EX: - Plaintext: HELLO WORLD

Ciphertext: LLOHE OWLRD

⇒ The same letters are still present. Therefore, an attacker can calculate letter frequencies and use structural pattern to help recover the plaintext.

⇒ Simple substitution also has weakness. Particularly because a fixed substitution preserves frequency relationship between plaintext and ciphertext.

Final Analysis :-

Technique

Vulnerability

Reason

Substitution

High for simple

Frequency Patterns

cipher

are preserved.

Transposition

High when used
alone

original character and
frequency remain unchanged

combination

lower

combining substitution +
transposition provides;
better confusion and
diffusion

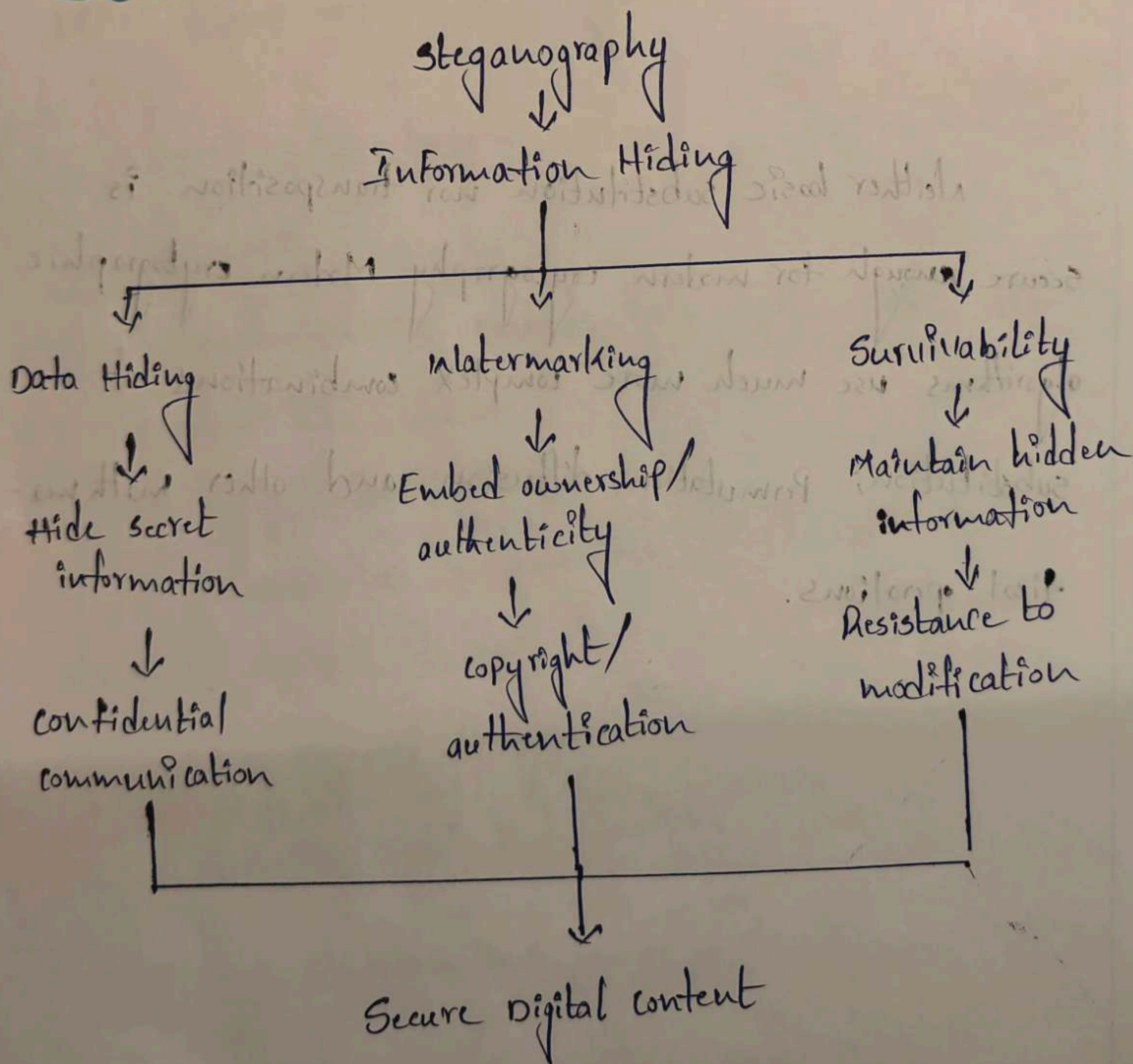
Conclusion :-

Neither basic substitution nor transposition is
secure enough for modern cryptography. Modern cryptographic
algorithms use much more complex combinations of
substitution, permutation, diffusion, and other mathema-
tical operations.

2, Steganography - Data Hiding, Watermarking and Survivability

To construct a concept map connecting data hiding, watermarking and survivability and analyze how each contribute to secure communication with real world applications

Concept Map



A) Relationships Between the Concepts

Data Hiding :-

data hiding means concealing information inside another medium, such as:

⇒ Image

⇒ Audio

⇒ Video

⇒ Text

EX! - secret message + Image



stego Image

Watermarking :-

watermarking embeds information such as:

⇒ owner identification

⇒ copyright information

⇒ Authentication information

into digital content

original Image



copy right Information



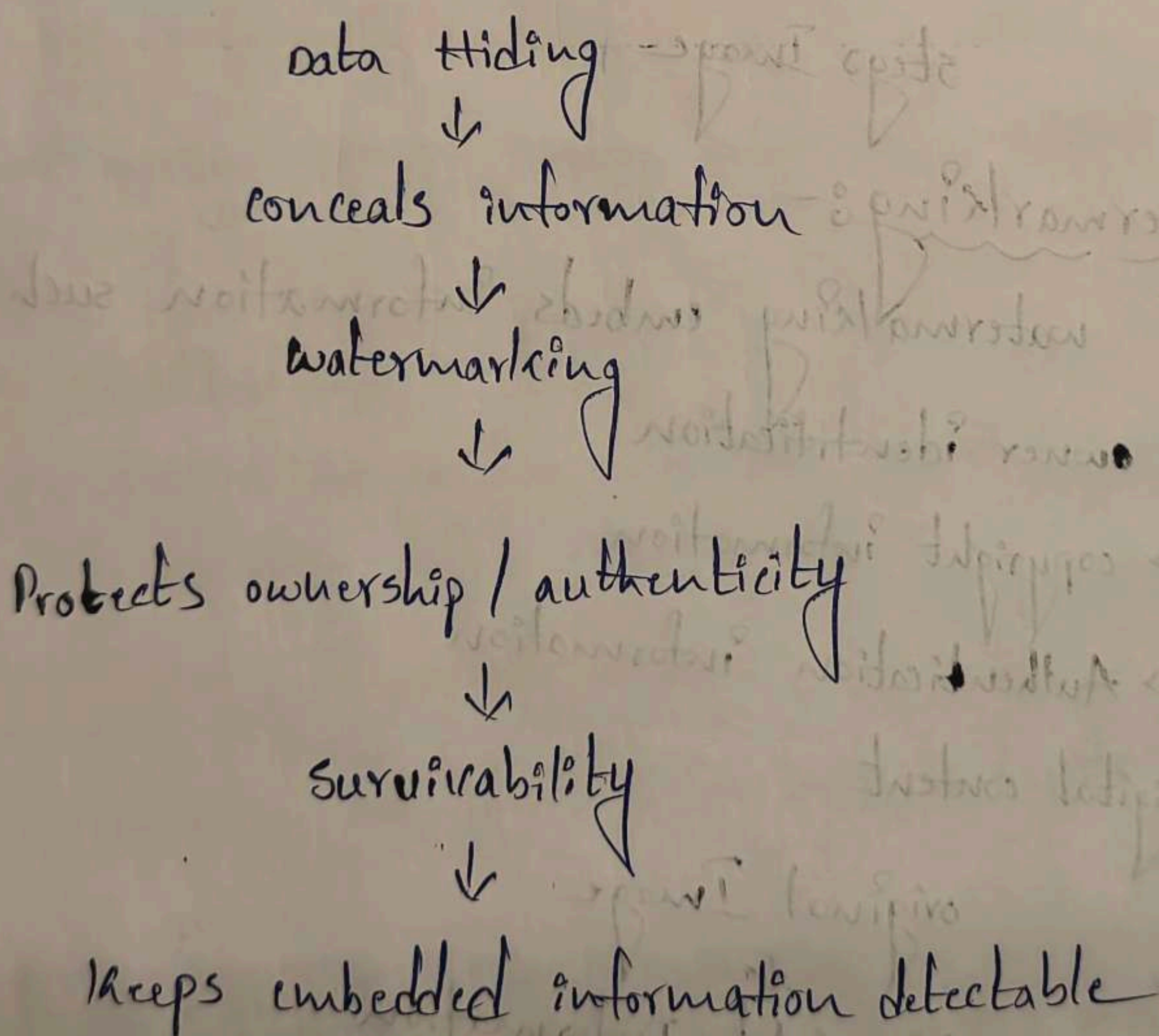
watermarked Image

Survivability :- Survivability refers to the ability of the embedded information to remain detectable after normal modifications

Examples of modifications

- compression
- Resizing
- Format conversion
- Minor editing

Relationship :-



B, contribution to Secure Communication

concept	contribution	Main purpose
Data Hiding	keeps secret information concealed	confidential communication
Watermarking	Establishes ownership/authenticity	copyright and authentication
Survivability	Maintains embedded information after modification	Reliable verification

Data Hiding → Confidentiality

A secret message can be hidden inside an innocent-looking image

Security contribution:

Makes the existence of the communication less obvious.

Watermarking → Authentication and Ownership

A watermarking can identify the legitimate owner of digital content.

Security contribution: - helps prove ownership and detect unauthorized use or manipulation.

Survivability → Reliability

A robust watermark should remain detectable after common operations such as compression.

Security contribution:

Ensure that authentication, or ownership information remains available.

→ Real-world Applications

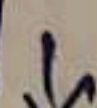
data hiding → secure communication

Application:- Secret message hidden inside an image

Secret data



LSB steganography



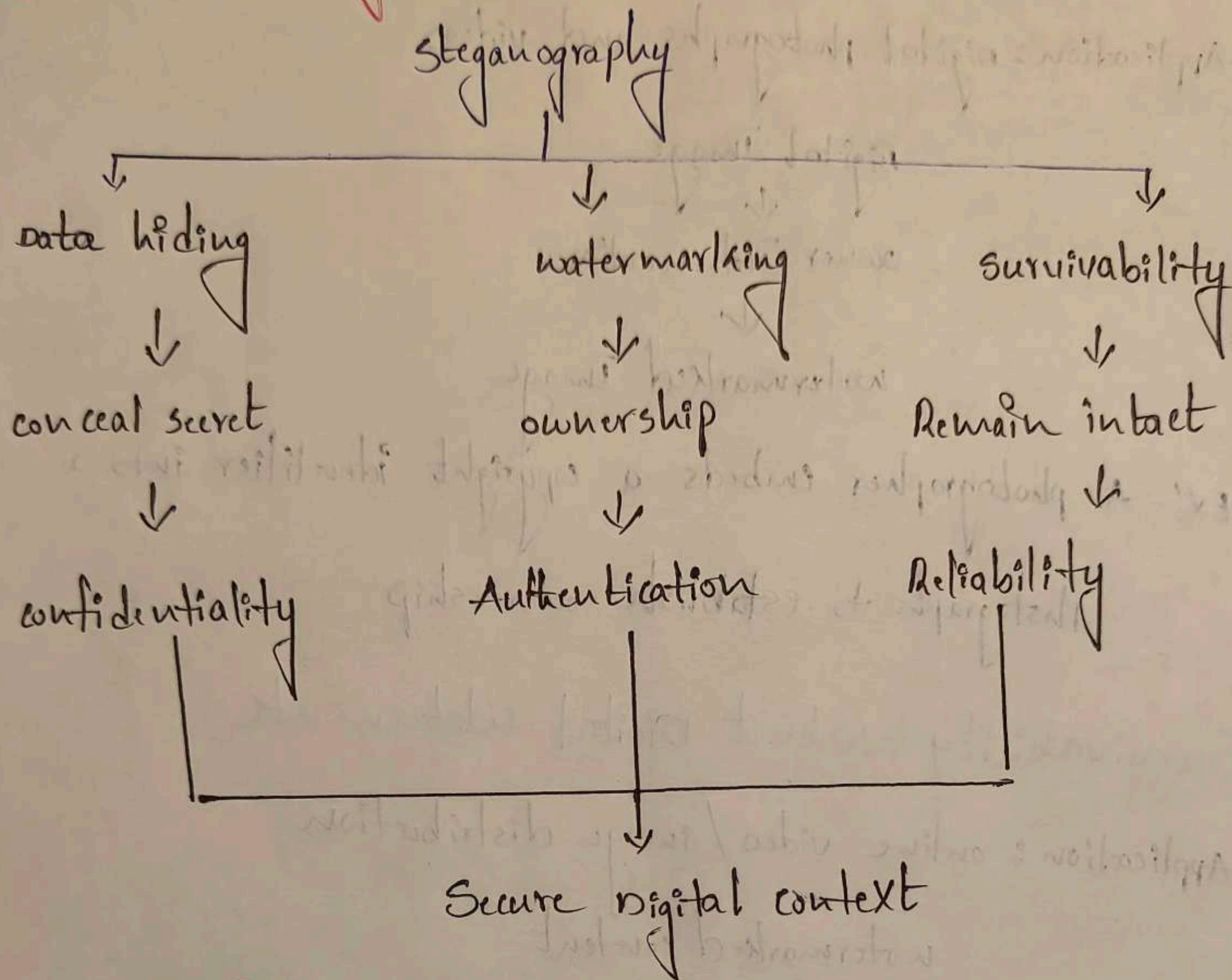
Digital Image



Receiver extracts secret data

Ex:- covert communication where the message is embedded inside an ordinary image

Final Analysis



Conclusion :-

- * Data Hiding :- Protects the secrecy of information
- * watermarking :- Protects ownership and support authentication
- * Survivability :- ensure that embedded information remains usable after modifications .

=> Together, they strengthen the confidentiality, authenticity and reliability of digital communication

Watermarking → copyright protection

Application: digital photographs and videos

digital image



owner information



watermarked image

EX! - A photographer embeds a copyright identifier into a

photograph to establish ownership

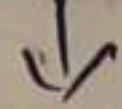
Survivability - Robust digital watermark

Application: online video / image distribution

watermarked content



compression



Resizing / Format conversion

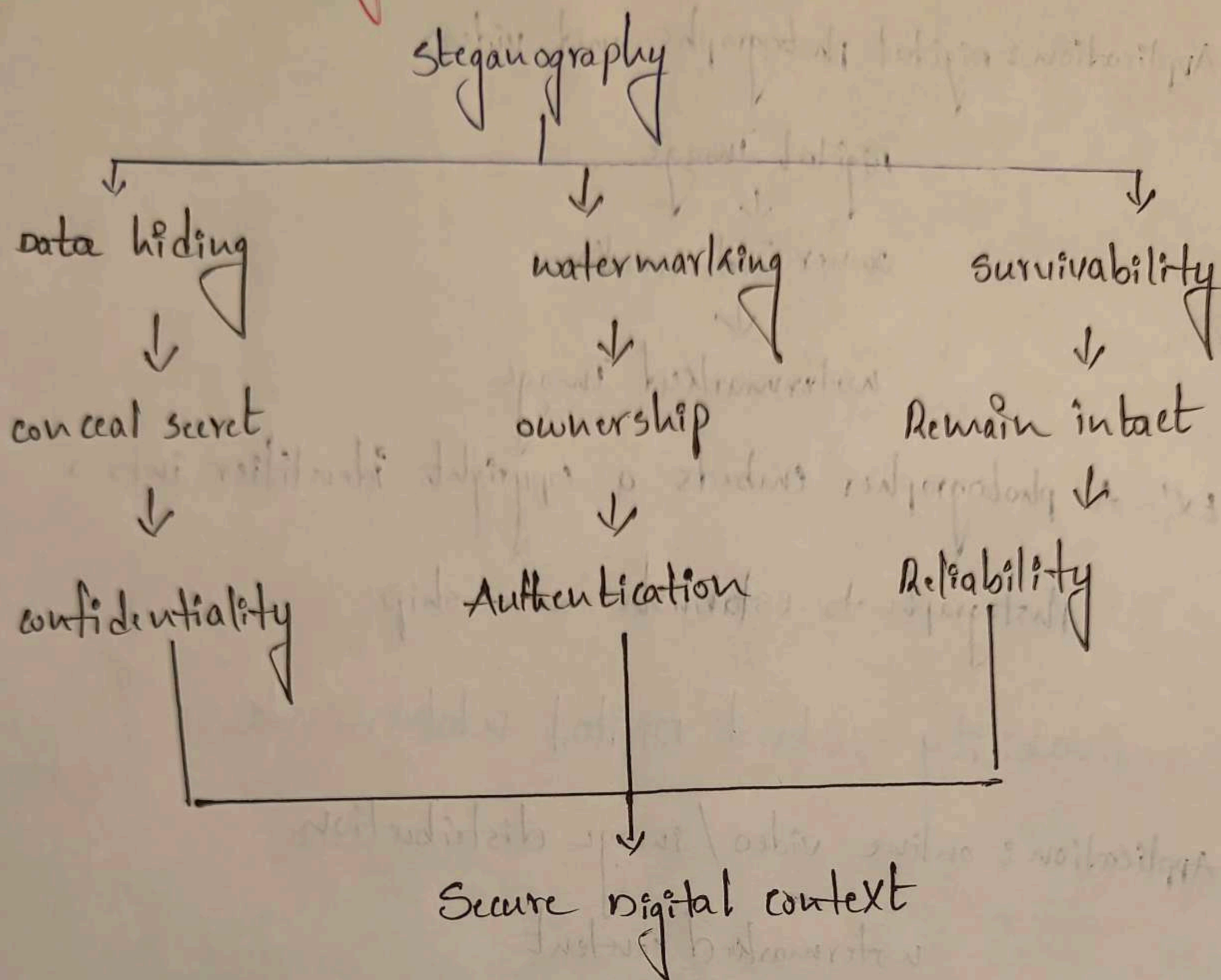


watermark still detectable

EX! - A video watermark remains detectable after online

compression, helping verify ownership.

Final Analysis



Conclusion :-

- * Data Hiding :- Protects the secrecy of information
- * watermarking :- Protects ownership and support authentication
- * Survivability :- ensure that embedded information remains usable after modifications .

=> Together, they strengthen the confidentiality, authenticity and reliability of digital communication